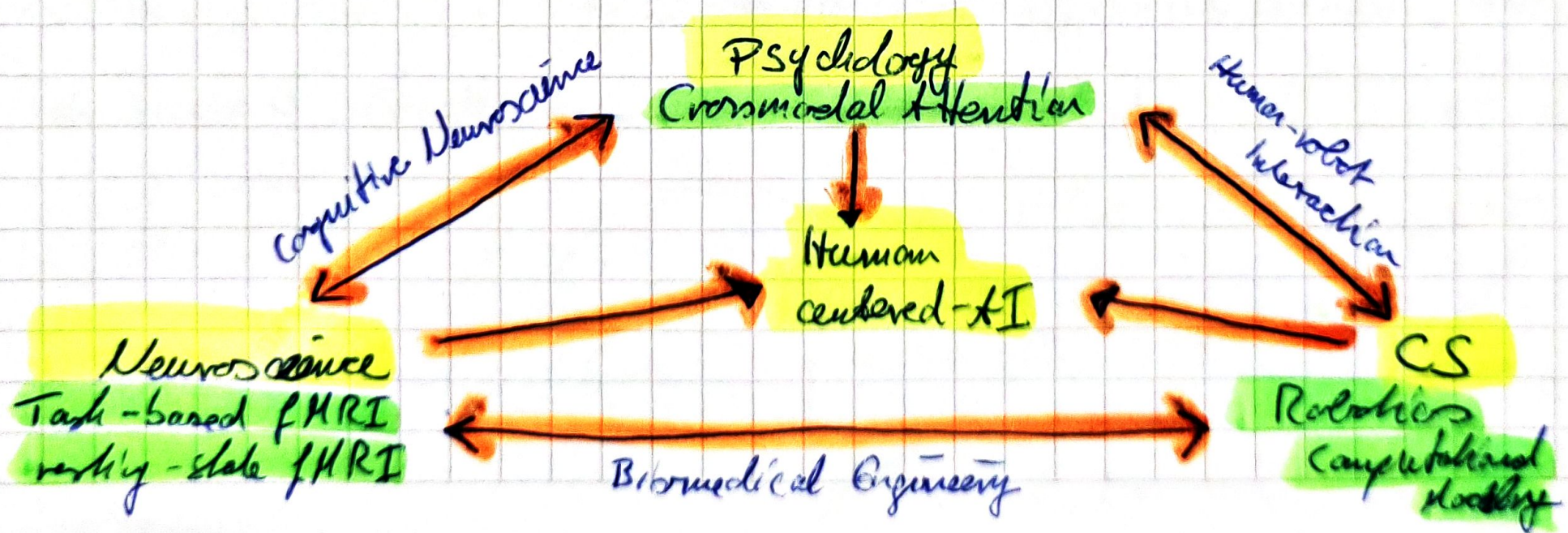


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Bio-inspired Attention

- Motivation: The art of being wise is the ability to know what to overlook



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Intro:

- Cognitive control: multiple processes that plan and coordinate actions to meet task goals
- Attention: the most important subfunction of cognitive control
- Selective Attention: like a filter with the ability to remove irrelevant information and thus optimize the current goal

Terminology:

- Selective Attention: (concentrate) (focus on a person in a noisy env)
 - Control awareness of the internal mind and ^{the} outside world
 - Integrate multidimensional and multimodal information
 - ↳ Ability to focus on specific stimuli while ignoring others (filter)
- Sustained Attention:
 - Maintain a state over time to detect the incoming stimulus
 - Enhance relevant stimuli and inhibit irrelevant distractors
 - ↳ over a longer period of time
 - ↳ focus on a single task without being easily distracted
- Exogenous Attention:
 - Bottom-up, stimulus-driven attention
 - Instinctive and spontaneous
 - ↳ External
- Endogenous Attention:
 - Top-down, goal-driven attention
 - Spotlight, allocate limited
 - ↳ Internal goals

maintain focus

allocate attention
→ correct on goal

Functional NN Model

- Alerting Network:
 - Ability to be alert for upcoming stimuli (Thalamus, frontal lobe)
- Orienting Network:
 - Focus on specific information among multiple sensory input (TPJ, SPL, FEF)
- Executive Control Network:
 - Monitoring and resolving conflicts between different inputs (ACC, dlPFC, AI)

perched (ACC)

• Measure Attentional Functions

→ **Reaction Time (RT)**: interval of time between the presentation of the stimulus and appearance of appropriate voluntary response in the subject

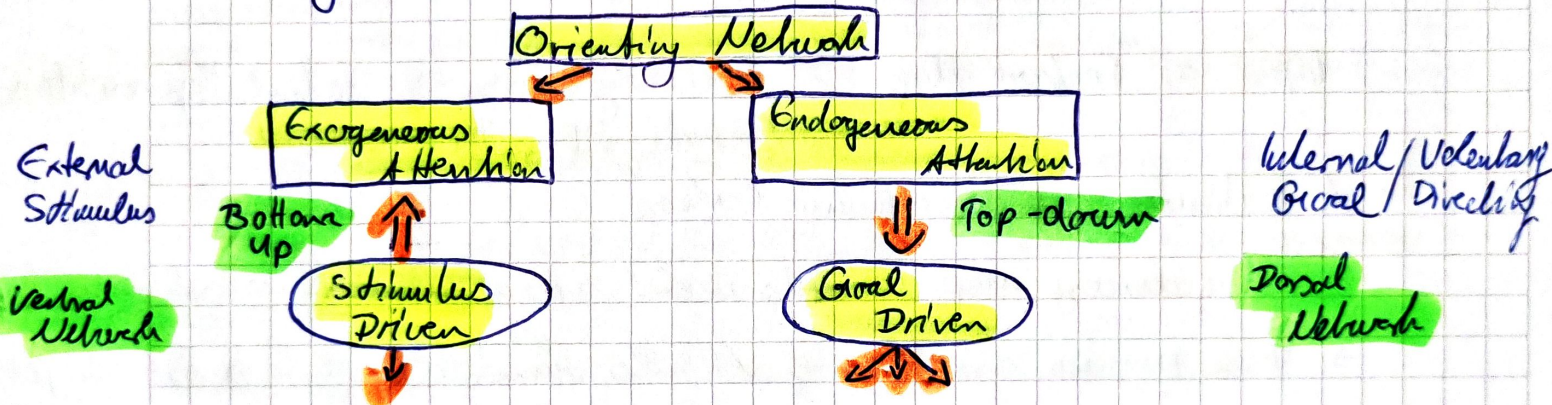
→ **Additive - factors method**:

→ procedure for analysing reaction time data to determine whether two variables affect the same or different processing stages

→ If two variables influence different stages, their effect should be additive

→ If two variables influence the same stage, their effect should be interactive

• Dichotomy between bottom-up (exo) and top-down (endo) control



• Cortical Oscillation Model

• Visual selective attention - "Pop-Out" Effect

→ Pop-Out happens when an object has more salient physical features than other objects in the context

→ **Physical Features**: location, color, shape, orientation, brightness, etc

→ Saliency could be extended to affective and social domain, like familiarity, threat, etc

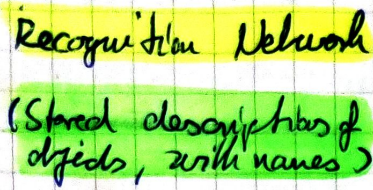
$$\text{Alerting Effect} = RT_{(\text{no cue})} - RT_{(\text{double cue})}$$

$$\text{Orienting Effect} = RT_{(\text{center cue})} - RT_{(\text{spatial cue})}$$

$$\text{Executive Control Effect} = RT_{(\text{incongruent})} - RT_{(\text{congruent})}$$

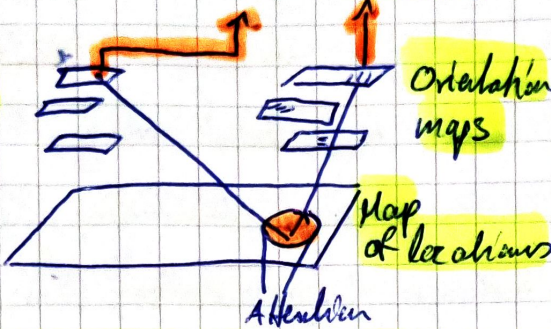
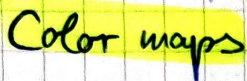
Attentional Network Test

- Feature integration theory (FIT) of attention



Temporary dyed Representation

(Time t , Place x)



Orientation maps

Map of locations

A. Henschen

Analyze
into features

Feature binding

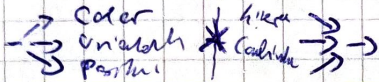
Object
↓
Pre-Heuristic Stage

Focused Attention Stage

↓ Wahrnehmung
Perception

* [Center - surround
differences are
nonlinear]

Model "Pop-Out" effect and saliency processing (Krupinski)



Input $\Rightarrow$ Feature Map $\Rightarrow$ Saliency Map $\Rightarrow$ Central Representation
* Winner-take-all is the core alg

* Winner-take-all is the core alg

- A saliency map in primary cortex

- The saliency map awards higher responses to more salient image locations
- Those responses are those of V1 cells tuned in input features (color, etc.)
- Firing rates of V1's output neurons increase monotonically with the saliency value of the visual input

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1u: - - 1 - -

Out: ---

Out 8 - - -

- * Winner - take all:

Only the most salient or active one is selected.

[Feature or region in an image
is identified. Ignore the others]

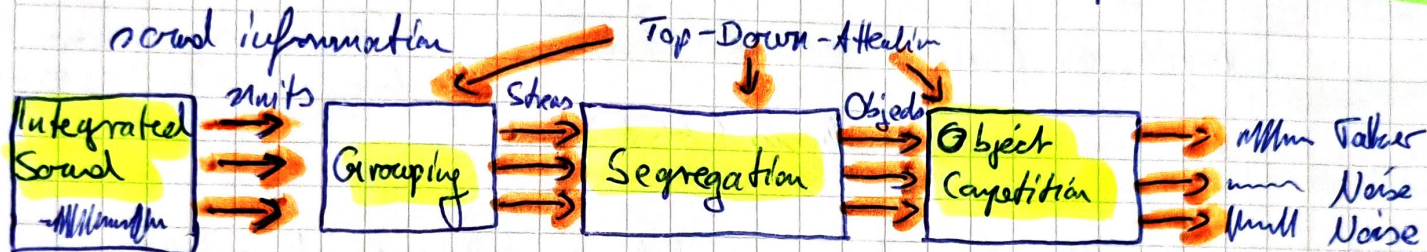
• Auditory Selective Attention - Cocktail Party Effect

→ Cocktail party effect:

→ At a noisy party, a person can concentrate on the target conversation (a top-down process) and easily respond to someone calling his/her name (a bottom-up process).

→ Auditory scene analysis (ASA) allows the auditory system to perceive and organize sound information from the environment.

→ Auditory attention: localize sound sources and filter out irrelevant sound information



→ Compound sound enters the bottom-up processing in the form of segregated features and then the features are grouped into streams

→ After segregation and competition, foreground sound stands out from the background noise

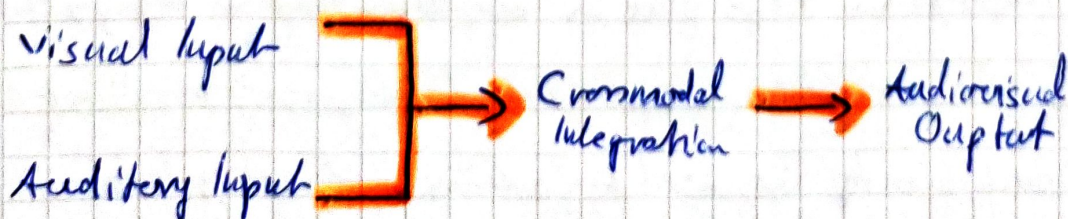
→ Top-down attention control can modulate processing on each stage

• Audiovisual crossmodal selective attention

→ Ventriiloquism Effect

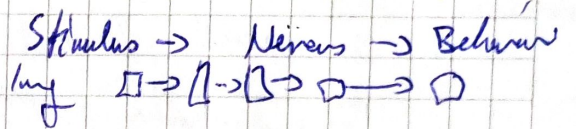
→ Auditory stimulus is perceptually shifted towards the position of the synchronous visual stimulus

→ Distance between audiovisual stimuli is crucial for sound localization



Take Home Message

- Concepts of selective, sustained, exogenous, endogenous attention
- Functions of Attention: alerting, orienting, and executive control
- Using reaction time (RT) and additive-factors method (Attentional Blink Test) to measure attentional functions
- bottom-up vs top-down
- Core algorithm of a saliency model: Winner-Take-All Strategy
- CNN are inspired by the hierarchical process of visual input on human's occipital lobe
- Auditory Scene Analysis (ASa)



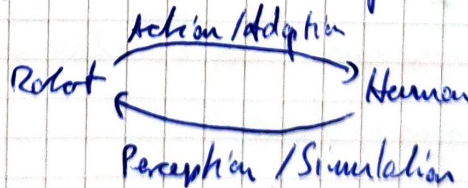
Social attention and crossmodal social cue integration

- Social attention helps human quickly learn how to interact with others, learn the language, and build social relationships

↳ Most crucial manifestation: ability to follow others eye gaze

- Can the Robot's Facial Expression and Gaze Impact on Human-Human-Robot Collaboration?

↳ Eye Connection & trust



- Human-Human-Robot Collaboration

H1: Actor
Arm Move

H2: Guide
Arm Move

Instruction: i: Cut
Gaze Shift, Verbal instructions

↳ facial expressions & eye gaze as completion time and ratings of the robot's intelligence

Robot happy face → faster completion, initial gaze to guide → more intelligent

- Gaze familiarity increased performance and perception

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• Social Attention in Human-Robot cooperation game

Human-Human Interaction: • human participants played the game with different personalities human assistants (introverted / extroverted / neutral)



Modeling Social Cues: • modeling social cues of human assistants into robots and designing a decision-making algorithm for the most suitable assistive action



Human-Robot Interaction: • human participants played with NAO robot assistants with different personalities (intro / extro) and different autonomy (WoT / fully)

• Modeling human-robot interaction

- verbal and non-verbal cues for both extroverted & introverted robots
- decision-making alg:
 - Personality-based
 - State-based

• Conclusion

- Human attention theories
- Psychological and neural mechanisms underlying visual, auditory, and audiovisual crossmodal selective attention
- Bio-inspired models, cognitive simulation, and human-robot interaction
- Comparison between human behaviour and modeling / robot performances